

Hany Hamed

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Researching generalization in reinforcement learning to build agents that adapt robustly to unseen, diverse settings

EDUCATION

University of Edinburgh (CDT-D2AIR), United Kingdom Sept 2026 — Sept 2030 (expected)

PhD in Artificial Intelligence for Robotics

Supervisor: Prof. Amos Storkey

Korea Advanced Institute of Science & Technology (KAIST), South Korea Aug 2022 — Aug 2024

Master of Science in Computer Science

Cumulative GPA: 3.95/4.30

Thesis Title: “Strategic Dreaming in MBRL: Leveraging State Space Structure for Sample-Efficient Generalist Agents”

Supervisor: Prof. Sungjin Ahn

Innopolis University, Russia Aug 2018 — Aug 2022

Bachelor of Computer Science (Robotics track)

Cumulative GPA: 4.68/5.00

Thesis Title: “Learning behavioral strategies for multi-robot system in a predator-prey environment”

Supervisor: Prof. Alexandr Klimchik & Consultant: Prof. Stefano Nolfi

EXPERIENCE

Generative Bionics

Research Engineer

Genoa, Italy

Jan 2026 — Sept 2026

- Developed vision-based humanoid reinforcement learning locomotion policies and sim-to-real deployment.
- Built procedurally generated simulation environments for realistic industrial scenarios.
- Reproduced and benchmarked recent research methods, integrating improvements into existing locomotion pipelines including whole-body policies.
- Contributed across the full RL pipeline, from data and training to validation, deployment, and live demos.

Artificial and Mechanical Intelligence Lab, IIT

Research Fellow (PI: Prof. Daniele Pucci & co-PI: Giulio Romualdi & Giuseppe L’Erario)

Genoa, Italy

March 2025 — Dec 2025

- Developed torque-control humanoid locomotion policies using reinforcement learning.
- Designed and implemented RL experiments for [ErgoCub](#) with IsaacLab.
- Built a Sim-to-Real transfer framework to enable robust humanoid locomotion.

University of Alberta & Amii

Research Assistant (PI: Prof. Rupam Mahmood (UoA) & Colin Bellinger (NRC))

Remote

Feb 2025 — Feb 2026

- Assisted in research on mask-based goal representation for robot learning, conducting experiments with multiple open-vocabulary object detection models to assess their effectiveness (Publications: [C3, C4]).
- Investigated model-based and model-free reinforcement learning algorithms for Sim-to-Sim transfer with finetuning in the target domain.

Machine Learning & Mind Lab (MLML), KAIST

Graduate Student Research Assistant (PI: Prof. Sungjin Ahn)

Daejeon, South Korea

Nov 2022 — Feb 2025

- Researched zero-shot task generalization in model-based RL, leading to a state-of-the-art agent (Publication: [C1]).
- Explored intrinsic motivation in hierarchical model-based RL, focusing on how high-level policies guide low-level policies.
- Worked on diffusion-based planning for long-horizon tasks while learning on short trajectories (Publication: [C2])

Unmanned Technology Laboratory, Innopolis University

Junior Developer (Industry Oriented)

Innopolis, Russia

June 2021 — June 2022

- Developed ROS packages for various drone modules, including gimbal control, RTK GPS, and payload control.
- Integrated Livox lidar with the lab’s drone for outdoor mapping.
- Developed a handheld Lidar device for indoor/outdoor mapping for an industrial partner ([results](#)).

Center of Robotics, Innopolis University

Undergraduate Research Assistant (PI: Prof. Sergei Savin)

Innopolis, Russia

Aug 2019 — April 2021

- Conducted RL experiments for a stabilizing control policy of a tensegrity hopper (Preprint: [P1]).
- Designed and implemented a contactless differentiable physics simulator for tensegrity robots using Taichi.
- Performed research on sim2real transfer for a three-prism tensegrity robot.

PUBLICATIONS

(C: conference / J: journal / P: preprint / *: equal contribution)

- [C4]] H. Hamed, A. Naik, C. Bellinger, R. Mahmood. “Rethinking the Suitability of Reinforcement Learning Algorithms Under Practical Transfer Constraints” In Reinforcement Learning Conference (RLC 2026) ([arXiv version](#)).
- [C3]] F. Shahriar, H. Wang, S. Alireza, G. Vasan, **H. Hamed**, A. Naik, C. Bellinger, R. Mahmood. “Dynamic Object Masks as Goal Representations for Visual Goal-Conditioned Reinforcement Learning.” In Reinforcement Learning Conference (RLC 2026) ([arXiv version](#)).
- [C2]] C. Chen*, **H.Hamed***, D. Baek, T. Kang, Y. Bengio, S. Ahn. “Extendable Planning via Multiscale Diffusion.” In AAAI 2026 (Oral). ([arXiv version](#))
- [C1]] **H. Hamed***, S. Kim*, D. Kim, J. Yoon, and S. Ahn. “Dr. Strategy: Model-Based Generalist Agents with Strategic Dreaming.” In International Conference on Machine Learning (ICML 2024).
- [J2]] G. Kulathunga, **H. Hamed**, and A. Klimchik. “Residual dynamics learning for trajectory tracking for multi-rotor aerial vehicles.” Scientific Reports (2024).
- [J1]] G. Kulathunga, **H. Hamed**, D. Devitt, and A. Klimchik. “Optimization-based trajectory tracking approach for multi-rotor aerial vehicles in unknown environments.” IEEE Robotics and Automation Letters (RA-L 2022).
- [P1]] V. Kurenkov, **H. Hamed**, and S. Savin, “Learning stabilizing control policies for a tensegrity hopper with augmented random search” ([arXiv version](#))

AWARDS

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| • UKRI AI CDT-D2AIR Fully Funded PhD Studentship | 2026-2030 |
| • ICML Travel Grant | 2024 |
| • KAIST Graduate School Full Scholarship (Master) | 2022-2024 |
| • Outstanding Achievements at Innopolis University | Fall 2020 |
| • Outstanding Contribution to Science at Innopolis University | 2020, 2021 |
| • 3rd place DOTS competition from Bristol Robotics Lab and Toshiba | June 2021 |
| • Innopolis University Full Scholarship (Bachelor) | 2018-2022 |

Teaching Experience

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| • TA in CS492 “Deep Reinforcement Learning”, KAIST | Spring 2023 |
| • TA in “Introduction to ROS” course, Innopolis University | Summer 2022 |

Academic Service

- **Conference Reviewer:** ICLR (2024-2026), IROS (2024), ACML (2024), Humanoids (2025), AAAI (2026)
- **Workshop Reviewer:** ICML AutoRL (2024)
- **Journal Reviewer:** IEEE RA-L (2024-2025)

EXTRACURRICULAR ACTIVITIES

Korea Science and Engineering Fair (KSEF) <i>Judge</i>	South Korea 2022, 2023
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- Conducted technical interviews with the participants.
- Judged the research reports and participants’ presentations.

World Robotics Olympiad (WRO) International Finals

<i>Judge</i>	2022 - 2024
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- Developed the game rules of the international competition for the “Advanced Robotics Challenge (ARC)” category.
- Served as a head judge for the Russian qualifications.
- Served as a judge for the international competition in Győr, Hungary.